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In Collaboration with

ROBERT GORDON UNIVERSITY ABERDEEN

A Multimodal Approach to Solar PV Fault Localisation and Rectification

Group 22 A Requirement Specification Document by

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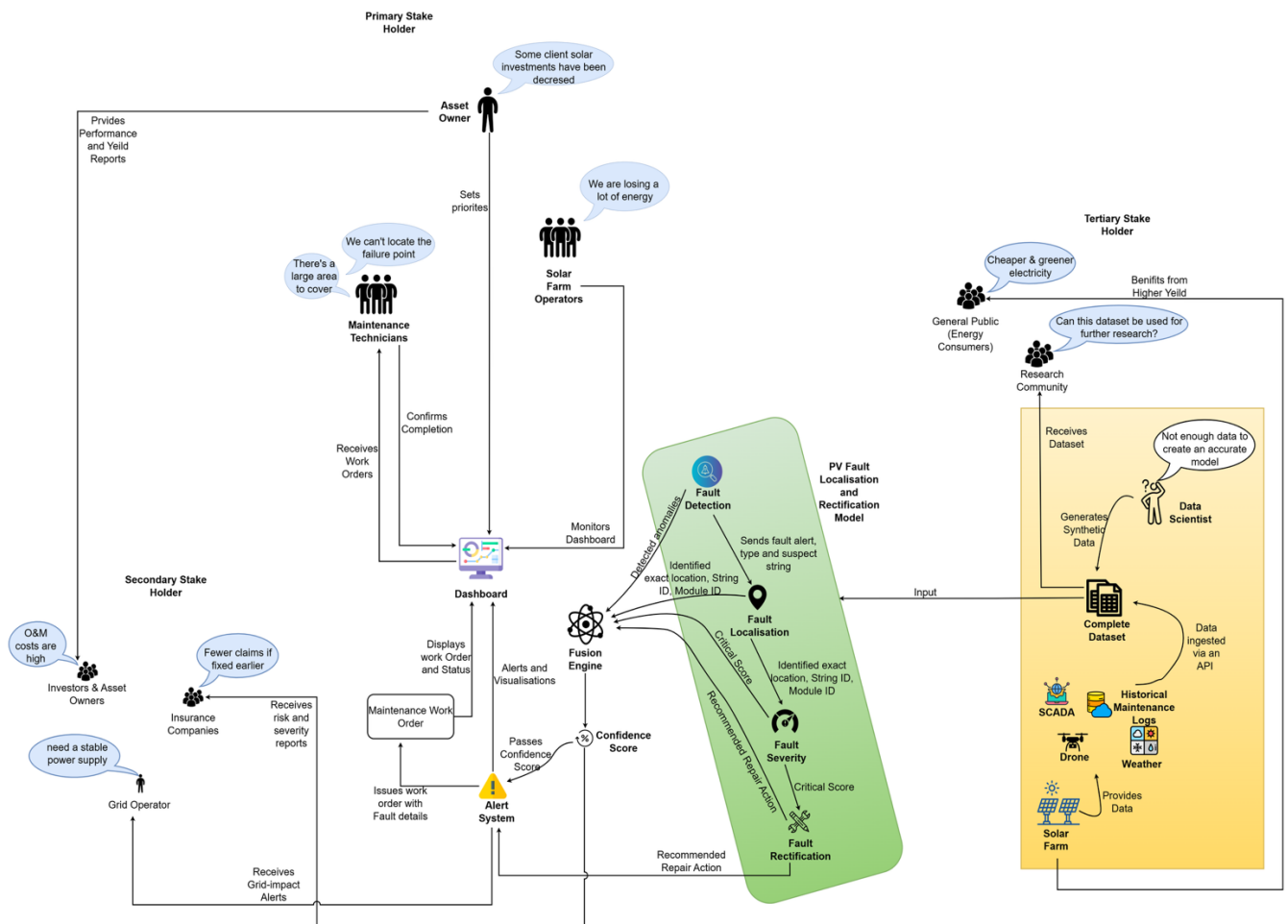
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1. Chapter Overview

This chapter details the systematic approach for gathering and analyzing requirements for the Solar Farm Fault Management System. It identifies the stakeholders using the Onion Model to categorize based on influence on the system. Various elicitation techniques, including literature analysis and surveys, are evaluated for their effectiveness in capturing both technical specifications and operational needs. The analysis is further supported by a use case diagram. Finally, the chapter defines the system scope through prioritized functional requirements covering the relevant components along with essential non-functional requirements.

2. Rich Picture

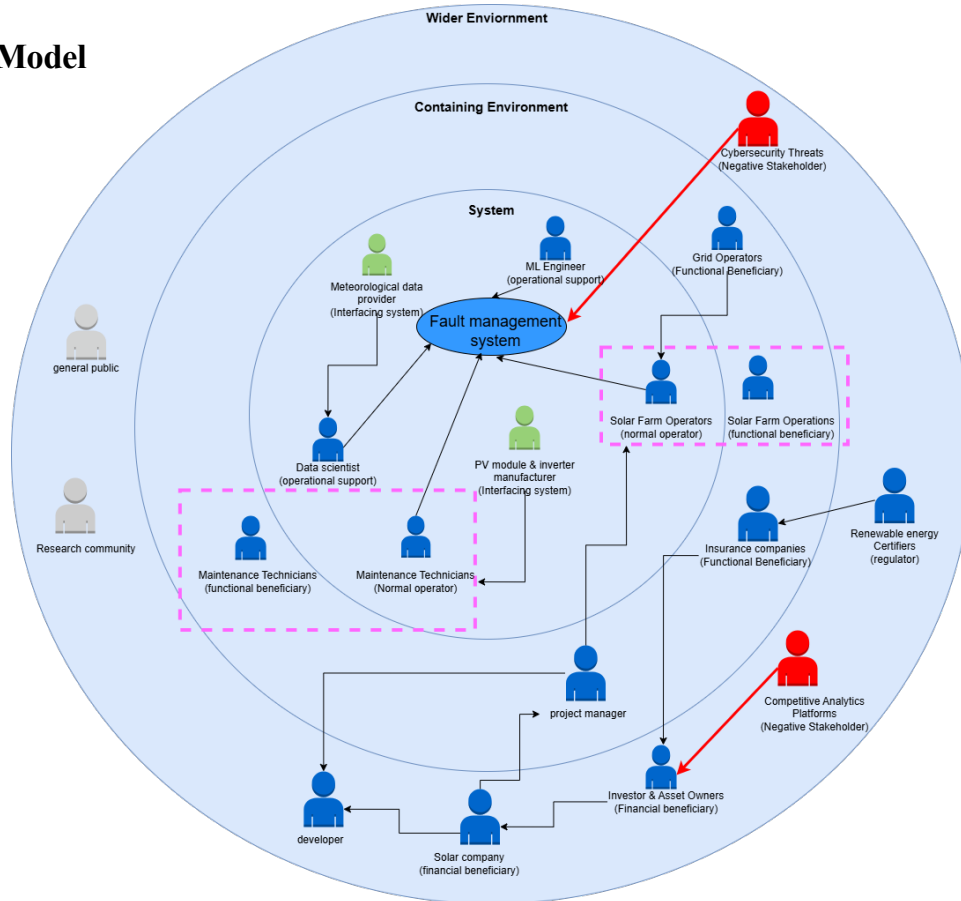


3. Stakeholder Analysis

Stakeholder Viewpoint

The Onion model implemented below maps all stakeholders influencing the Solar Farm Fault Management System. Their layered placement indicates their influence level, from normal operator to external influencer revealing the system requirements.

3.1 Onion Model



Stakeholder	Role	Viewpoints
Data Scientist	Operational Support	Gets clean real-time solar farm data and develops an accurate model to predict equipment failures before they happen.
ML Engineer	Operational Support	Implements and maintains production AI systems that automatically find and diagnose the problem.
Maintenance Technicians	Normal Operator/ (Functional Beneficiary)	Uses the system's diagnostic information for repairs and benefits from precise fault localisation, making their job faster and more efficient.
Solar Farm Operator	Normal Operator/ (Functional Beneficiary)	Operate the system and benefit from early fault detection that prevents costly downtime and equipment damage.
Meteorological Data Provider	Interfacing System	Provides weather data services to solar operations to help predict panel performance.
PV Module & Inverter Manufacturer	Interfacing System	Receives failure analytics and reports that aid in building improved products.
Grid Operator	Functional Beneficiary	Receives production forecasts of the solar farm, helping keep the electricity grid stable.
Insurance Companies	Functional Beneficiary	Uses risk data to reduce claims and offer better premiums.
Project Manager	Manages the System	Delivers projects that are quality standard and ensuring they are built on time with a possible budget.
Renewable Energy Certifiers	Regulator	Validate compliance and maintain industry standards.

Investor & Asset Owners	Financial Beneficiary	Protect investment with reliable returns and asset value.
Solar Company	Financial Beneficiary	Maximizes return on investment (ROI) through increased uptime and reduced costs.
Developer	Develops the System	Builds scalable software for critical energy infrastructure funded by the company.
General Public	Indirect Beneficiary	Prefers affordable, reliable, clean energy and benefits from lower electricity costs and more stable solar operations.
Research Community	Educational Beneficiary	Uses operational data for academic research.
Competitive Analytics Platforms	Negative Stakeholder	Attempts to steal performance data to gain a business advantage.
Cybersecurity Threats	Negative Stakeholder	Exploits system vulnerabilities for data theft or disruption.

4. Selection of Requirement Elicitation Techniques/Methods

This section evaluates requirement elicitation techniques for gathering system needs. Multiple methods are analyzed, highlighting their respective advantages and limitations.

4.1 Observing Current Systems and LR

This elicitation activity involved a literature review of existing solar farm monitoring platforms and academic research on PV fault diagnostics. The analysis aimed to benchmark current capabilities and identify critical gaps in detection accuracy, localization precision, and automated rectification strategies.

Advantages	Disadvantages
It helps identify proven algorithms used in the domain, such as LSTM/CNN models for time-series fault detection and gradient boosting for severity classification.	Interpreting cutting-edge academic research on PV fault diagnostics is often challenging due to complex methodologies and a significant gap between theoretical models and field deployment.
It exposes specific shortcomings in existing solutions, for instance, a common lack of precise, module-level fault localization or insufficient integration of real-time meteorological data for accurate severity analysis.	The inferences drawn from literature require substantial interpretation to be contextualized for a specific location, such as adapting fault models for the unique climate, dust profiles, and common equipment used in a particular region.

4.2 Surveys and Questionnaires

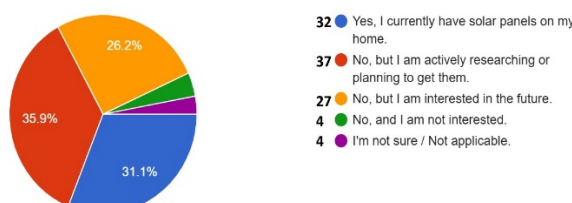
For a solar farm fault management system, surveys are a practical method to gather input from the public as well as a distributed group of stakeholders.

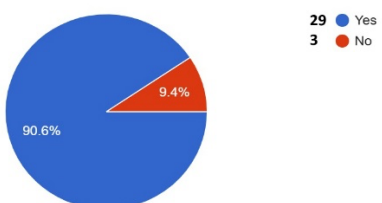
Advantages	Disadvantages
Efficiently collects feedback from both technical stakeholders and the public on issues like system reliability and community impact.	Technical questions may be misunderstood by public respondents, while stakeholders may oversimplify complex operational needs.

Cost-effective for gathering standardized data across different geographic locations and stakeholder groups.

Varying response formats between public and technical stakeholders can create integration challenges for data analysis.

4.3 Discussion of Results

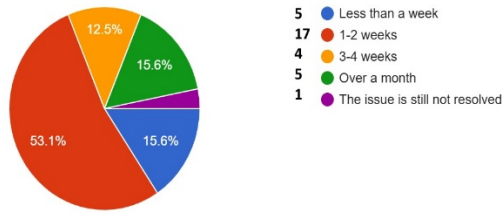
Question	Which of the following best describes your current status with solar panels?
Aim of Question	It identifies current users, potential customers, and uninterested parties to reveal specific needs and barriers for each group to inform system requirements.
Observations 103 responses  32 ● Yes, I currently have solar panels on my home. 37 ● No, but I am actively researching or planning to get them. 27 ● No, but I am interested in the future. 4 ● No, and I am not interested. 4 ● I'm not sure / Not applicable.	
Conclusion - The results highlight a critical gap between public interest in solar energy and actual adoption rates. While a strong majority (67%) express interest or planning, the low current ownership (26.2%) suggests significant barriers prevent conversion. This underscores the need for a solar farm fault management system that provides irrefutable, data-driven proof of reliability and ROI. By offering automated fault detection and performance analytics, the system mitigates perceived risks, empowers informed decision-making for potential customers, and ultimately helps the solar industry bridge this adoption gap.	

Question	Have you ever experienced a fault or issue with your solar panel system?
Aim of Question	To measure the real-world reliability of residential solar panels and identify how common system faults are among current owners.
Observations 32 responses  29 ● Yes 3 ● No	
Conclusion - The extremely high incidence of faults (90.6%) demonstrates that system reliability is a critical and nearly universal pain point for solar owners. This data strongly validates the core need for a fault management system. It shifts the problem from a rare occurrence to a fundamental operational challenge, justifying the development of a predictive and responsive diagnostic tool to enhance customer satisfaction and trust in solar technology.	

Question	On average, how long did it take from noticing a problem to having it fully resolved?
Aim of Question	To measure the efficiency of the current solar panel repair and maintenance process and gauge customer downtime.

Observations

32 responses

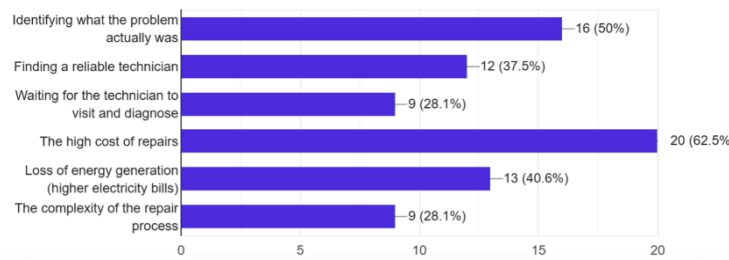


Conclusion - While most issues (53.1%) are resolved very quickly between one and two weeks, a surprising percentage of users (31.25%) experienced very long delays, taking over a month or reporting that the issue was still not resolved.

This indicates a service delivery model that is efficient for most cases but struggles with resolving more complex or persistent issues, leading to a considerable segment of users facing extended wait times.

Question	How much did it typically cost to diagnose and repair a fault? (Please include any service call fees)
Aim of Question	To identify the primary pain points and bottlenecks customers face during the solar panel repair process.

Observations

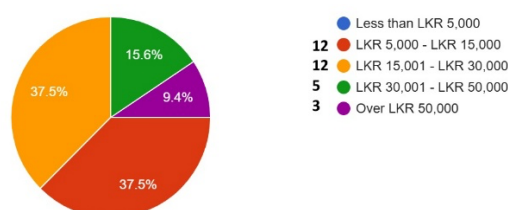


Conclusion - The results pinpoint the core failures in the current fault resolution ecosystem. The overwhelming challenge is cost (62.5%), followed by the fundamental inability to identify the problem (50%). This directly validates the need for an intelligent fault management system that provides immediate, accurate diagnostics to reduce uncertainty, control costs, and streamline the entire repair workflow.

Question	How much did it typically cost to diagnose and repair a fault? (Please include any service call fees)
Aim of Question	To quantify the financial burden of solar panel repairs on customers and identify common cost brackets.

Observations

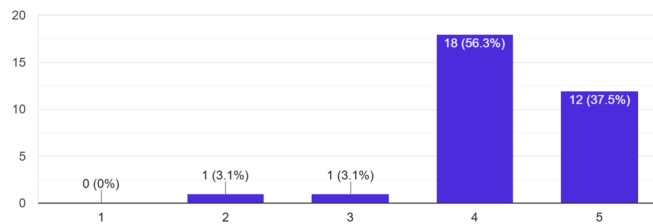
32 responses



Conclusion - The cost for repairs is highly polarized, which created two distinct customer experiences. A significant 37.5% of users paid minimal costs (under LKR 5,000), while an equal 37.5% faced very high expenses (over LKR 50,000). This contrast suggests that issues are typically either simple, low-cost fixes or require major, expensive repairs, with very few customers falling into a moderate cost range.

Question	How valuable would a service be that could automatically detect faults, tell you exactly what's wrong, where it is, and how to fix it?
Aim of Question	To validate market demand and perceived value for the proposed automated solar fault management system.

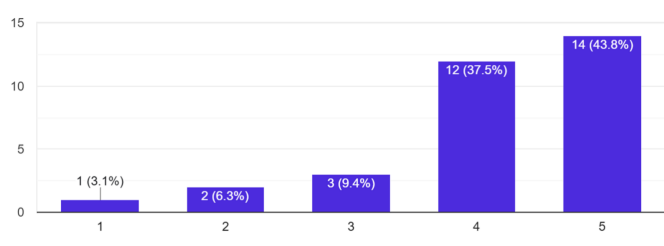
Observations



Conclusion - The response is overwhelmingly conclusive. An exceptional 93.8% of respondents (56.3% + 37.5%) find high value in an automated diagnostic service. This demonstrates a clear and urgent market need for a solution that eliminates the guesswork and delays currently plaguing solar panel maintenance. The data provides direct validation that the core functionality of the proposed system automating detection, localization, and repair guidance is precisely what customers are demanding to solve their most critical pain points.

Question	How much do you agree that having an AI fault management system would have made you install solar sooner?
Aim of Question	To measure if an AI fault management system acts as a key incentive that can accelerate solar adoption decisions.

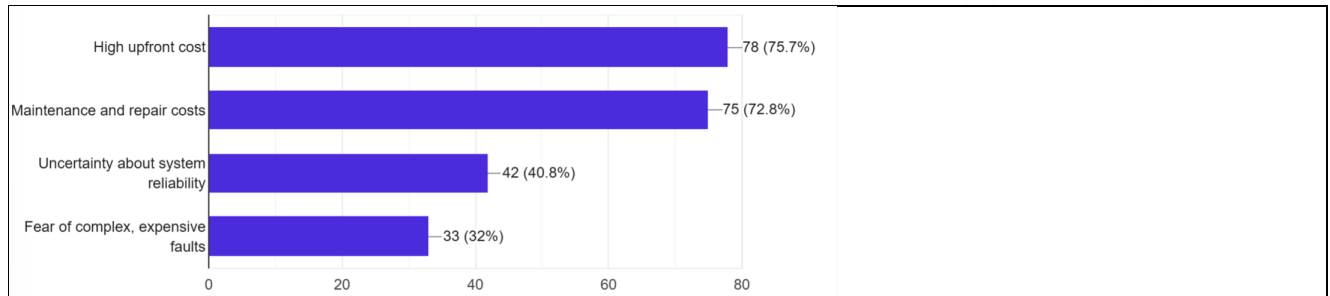
Observations



Conclusion - The results confirm that system reliability is a major factor in adoption timing. A strong majority (81.3%) agree that an AI management tool would have influenced them to install solar sooner. This proves that perceived risk and maintenance concerns are significant barriers delaying market growth. The proposed system is not just a diagnostic tool but a powerful enabler that can boost consumer confidence and accelerate the transition to solar energy.

Question	What is your biggest concern about installing solar panels on your home? (Select up to two)
Aim of Question	To identify the primary psychological and financial barriers preventing potential customers from adopting solar technology.

Observations

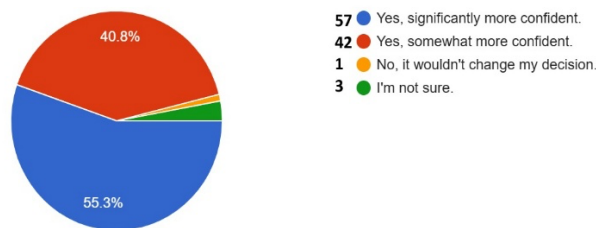


Conclusion - The primary barrier to adopting home solar panels is overwhelmingly financial. Most respondents (75.7%) are concerned about the high upfront cost, while a significant 72.8% are worried about ongoing maintenance and repair costs. These two financial concerns far outweigh other issues like uncertainty about system reliability (40.8%) or fear of complex faults (32%), indicating that cost is the central obstacle for potential customers.

Question	Would a system that proactively monitors for faults and provides clear, instant diagnostics make you feel more confident about investing in solar?
Aim of Question	To test if an automated fault monitoring system can effectively build consumer trust and mitigate the primary fear of repairs.

Observations

103 responses

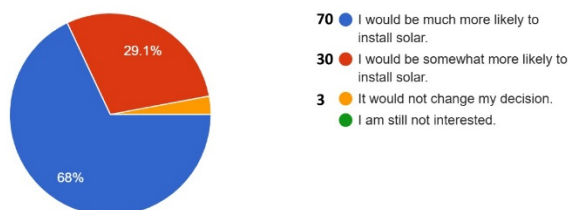


Conclusion - The results are clear: 96.1% of respondents say a proactive monitoring system would boost their confidence in investing in solar. This shows the fault management system not only diagnoses issues but also reduces perceived risk, reshapes market perception, and drives demand by providing essential reassurance to potential customers.

Question	If an AI system could significantly reduce both the initial cost and the long-term maintenance costs of solar, how would this affect your decision?
Aim of Question	To quantify how much AI-driven cost reduction in both installation and maintenance would accelerate solar adoption.

Observations

103 responses



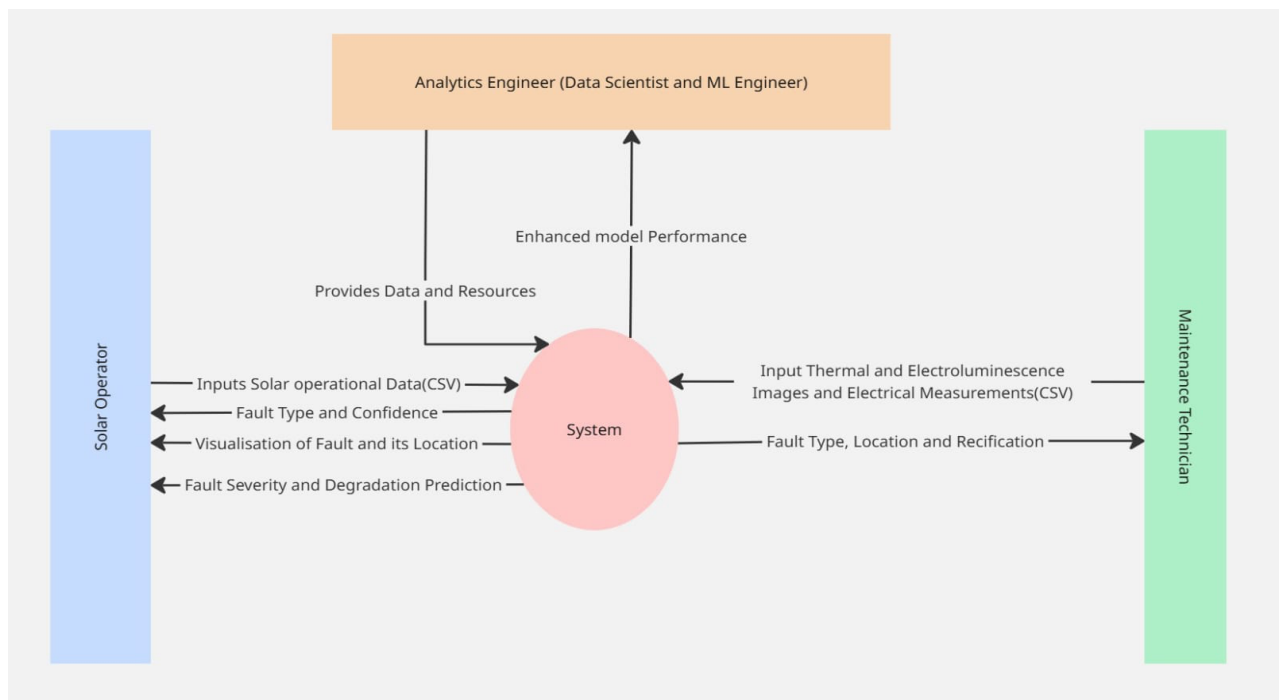
Conclusion - The response is overwhelmingly positive, with a massive 97.1% of potential customers indicating they would be more likely to install solar. This proves that cost is the fundamental barrier, and an AI solution that directly tackles both upfront and long-term expenses is the most powerful catalyst for market transformation. The data validates that integrating cost-

reduction features is not just an enhancement, but a core requirement for achieving widespread solar adoption.

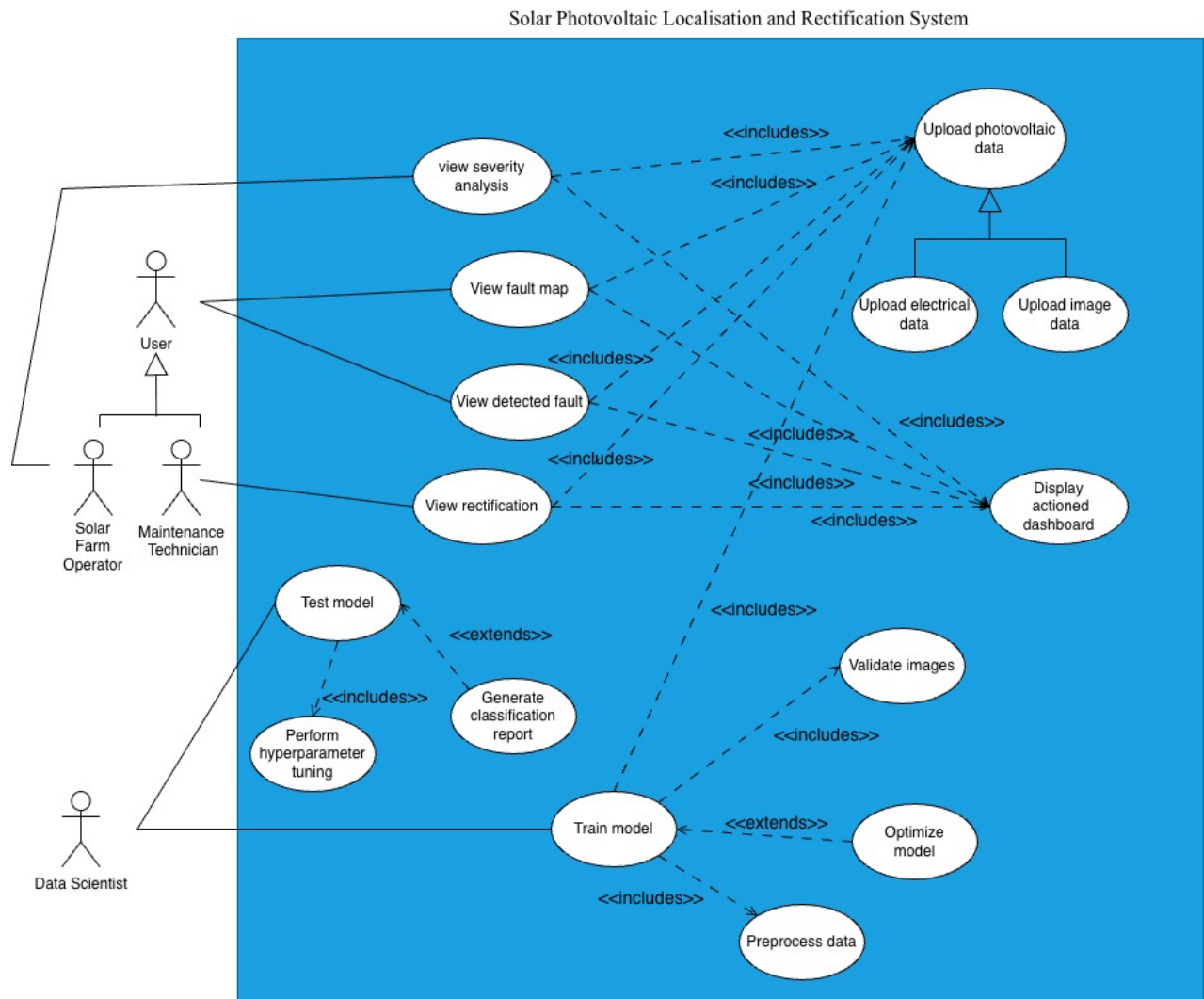
4.4 Summary of Findings

Findings	LR	Questionnaire
Current PV monitoring systems lack precise, module-level fault localization capabilities	X	X
Maintenance technicians face challenges with inaccurate fault diagnosis leading to multiple site visits		X
Limited integration of real-time meteorological data affects fault prediction accuracy	X	
Strong industry demand for AI-driven predictive maintenance and automated severity analysis		X
Manual inspection and diagnostic processes remain time-consuming and inefficient	X	X
Existing commercial solutions show poor integration between fault detection and maintenance workflows	X	
Need for automated maintenance work order generation and repair tracking systems		X
Grid operators require accurate power forecasts and early outage predictions for stability		X
Insurance companies need reliable performance data for risk assessment and premium calculation		X

5. Context Diagram



6. Use Case Diagram



Use Case Descriptions

Use Case Name	Train model
Section	Main
Actor(s)	Data Scientist
Overview	Explains how the data scientist will train machine learning models using historical fault data and electrical measurements to improve detection accuracy.
Purpose	To build and refine models that will enable automation for the system to accurately detect, localise, classify and rectify solar PV faults.
Pre-Conditions	Sufficient data is available, and the Data Scientist is authorized and has access to the system.
Post-Conditions	The models are trained and available for testing and further development.
Main Flow of Events	1. The data scientist initiates the process by accessing the interface to train the models. 2. The system shows available datasets for training. 3. The data scientist selects algorithms with their training parameters. 4. The system processes the data and trains the model. 5. The data scientist monitors the model's performance metrics. 6. The system saves the trained model for future use.
Alternative Flow	1. If the model's performance is poor, the system suggests alternative algorithms or parameter tuning. 2. If saving the model fails, the system notifies the data scientist to handle the issue.
Includes	Upload photovoltaic data, Validate images, Preprocess Data
Extends	Optimize Model

Use Case Name	Train model
Section	Main
Actor(s)	Data Scientist
Overview	Explains how the data scientist will test and evaluate the performance of the models trained using validation/testing datasets along with performance metrics.
Purpose	To ensure that the models meet the accuracy requirements before being deployed into the system.
Pre-Conditions	Sufficient validation data is available, and the models have been trained.
Post-Conditions	The models have been successfully validated and may be deployed into the system.
Main Flow of Events	1. The data scientist initiates by selecting a trained model for testing. 2. The data scientist then loads the validation dataset into the model. 3. The system will validate the model. 4. The system will optionally generate performance metrics scores 5. The data scientist performs hyperparameter tuning while testing the model
Alternative Flow	If test data contains missing values or pre-processing of the

	dataset incomplete, the testing stops, and the user is prompted to use a valid validation dataset.
Includes	Perform hyperparameter tuning
Extends	Generate classification report

Use Case Name	View detected fault
Section	Main
Actor(s)	User
Overview	Explains how the user views the types of faults detected by the system, which includes confidence scores and fault classifications in the solar module/string using imagery and textual data
Purpose	To alert the user an overview of all the detected faults
Pre-Conditions	The system has processed image and textual data.
Post-Conditions	The fault(s) were successfully detected along with their confidence scores.
Main Flow of Events	1. The user accesses the system dashboard 2. The system will display the set of detected faults with their types along with their confidence scores 3. The user can filter by type 4. The system then shows detailed information (such as severity) for each fault 5. The system also shows the relevant confidence scores for the respective faults for reliability.
Alternative Flow	If no faults were detected, then the system informs the user that no faults are present
Includes	Upload photovoltaic data, Display actioned dashboard
Extends	None

Use Case Name	View fault map
Section	Main
Actor(s)	User
Overview	Explains how the user will view an interactive map showing precise locations of detected faults within the solar farm.
Purpose	To allow the user to visually locate faults on the solar farm.
Pre-Conditions	A fault has been detected in the photovoltaic module.
Post-Conditions	A visualized interactive fault map is displayed to the user.
Main Flow of Events	1. The user initiates by selecting the fault map. 2. The system will process the user's choice. 3. The system will then display an interactive fault map of the solar farm, showing the detected fault and its location. 4. The system will also additionally display colored markers for the respective fault and severity 5. The system will also show string and module numbers for each fault.
Alternative Flow	If the fault map failed to be displayed, then the user is given the string and module numbers of the faults while also being notified to restart the web application for the complete fault map.
Includes	Upload photovoltaic data, Display actioned dashboard
Extends	None

Use Case Name	View severity analysis
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Section	Main
Actor(s)	Solar Farm Operator
Overview	Explains how the solar operator views the assessment of detected faults, which includes degradation levels, cost repairs and power loss.
Purpose	To help the operators prioritise fault repairs based on their severity and cost/power cost.
Pre-Conditions	The faults have been detected in the solar module
Post-Conditions	The operator is shown a severity analysis for the detected faults for immediate attention.
Main Flow of Events	1. The operator accesses to view the severity analysis 2. The systems display faults categorised by severity level (Low/Medium/High/Critical) 3. The system will also show power loss and revenue impact for each fault 4. The system additionally displays historical trends and degradation patterns 5. The operator uses filters to sort by severity, impact or re- pair urgency 6. The system will then provide visual charts showing severity distribution
Alternative Flow	If the visual charts fail to be displayed due to an internal error, the operator is displayed a basic severity classification for each fault while also requesting them to restart the application for the full analysis.
Includes	Upload photovoltaic data, Display actioned dashboard
Extends	None

Use Case Name	View rectification
Section	Main
Actor(s)	Maintenance Technician
Overview	Explains how detailed maintenance repair recommendations will be viewed by the technicians
Purpose	To provide the maintenance technicians with actionable repair guidance, including steps, required tools, and safety precautions.
Pre-Conditions	Fault(s) have been detected, localised and severity analysis is complete.
Post-Conditions	The technician has clear instructions to perform the repair and necessary information about the required tools.
Main Flow of Events	1. The technician selects a specific fault 2. The system displays the recommended rectification action 3. The technician views the step-by-step repair procedures 4. The system displays the required tools, steps and safety precautions.
Alternative Flow	If multiple repairs exist, then the system shows alternatives that are more cost-effective.
Includes	Upload photovoltaic data, Display actioned dashboard
Extends	None

7. Functional Requirements (with prioritization)

Priority Level Description

- **High** - Mandatory Requirements
- **Medium** - Non-Mandatory Requirements
- **Low** - Out of Scope Requirements

FR No.	Functional Requirements and Descriptions	Priority
FR1	Electrical data (CSV) and image data is sent to the model	High
FR2	Detection of faults in the PV System	High
FR3	Classification of the fault types and their confidence	High
FR4	PV Array Segmentation into strings and modules	High
FR5	Use of fault localisation Fusion Engine to determine the fault location using confidences	High
FR6	Fault Severity Classification and Degradation Prediction	High
FR7	Fault Severity Data Fusion Engine for accurate classification	Medium
FR8	Rectification Recommendation for identified faults and provide better interpretation of the steps to rectify the fault	High
FR9	Report Generation of Fault Analysis	Medium
FR10	User Interface Display	High
FR11	User Feedback Input and clarification	Low
FR12	The solar operator uses the analysis to observe faults and the performance of the solar panels	High
FR13	The maintenance technician uses the rectification results to perform the recommended restorations	High

8 Non-Functional Requirements

Accuracy – Critical for fault detection, localisation and severity assessment to ensure correct diagnosis and prevent unnecessary maintenance costs, system downtime in solar PV operations.

Performance – Must process large volumes of electrical and image data efficiently to ensure fast identification of faults and minimise energy loss in solar installations.

Security – Implements strong data protection mechanisms to safeguard proprietary operational data, image data and system performance metrics from unauthorised access

Usability – Provides intuitive dashboards and clear visualisations that enable operators, technicians and managers of all technical levels to quickly understand fault reports.

Maintainability – Designed for fast updates, retrain with new data and straightforward maintenance enabling quick deployment of improvements and minimal downtime.

9 Requirements and Description

ID	Description	Specification	Priority
NFR01	The system must accurately detect, localise, assess severity and recommend rectification actions for PV faults.	Accuracy	High
NFR02	The system must process electrical and image data within acceptable time limits.	Performance	Medium
NFR03	The system must process fault detection and severity analysis within an acceptable time response.	Performance	Medium
NFR04	The system should run efficiently on standard mid-range hardware.	Performance	Medium

NFR05	All data must be protected from unauthorized access during upload, processing, and storage.	Security	High
NFR06	The user interface should be simple and intuitive for solar technicians and operators.	Usability	Medium
NFR07	The system must support increasing data volumes and large PV farms.	Scalability	Medium
NFR08	The system must be easy to maintain and update.	Maintainability	Medium

10 Chapter Summary

This chapter presents a comprehensive requirements analysis for a Multimodal Solar PV Fault Localisation and Rectification System. Using the Onion Model, stakeholders from operators to regulators were mapped. Literature reviews, surveys revealed critical gaps in existing systems, particularly in module-level fault localisation and predictive maintenance. Key findings informed prioritized functional requirements (fault detection, localisation, severity analysis and rectification recommendations) and essential non-functional aspects, supported by a rich picture, context diagram, use case diagram and detailed use case descriptions.